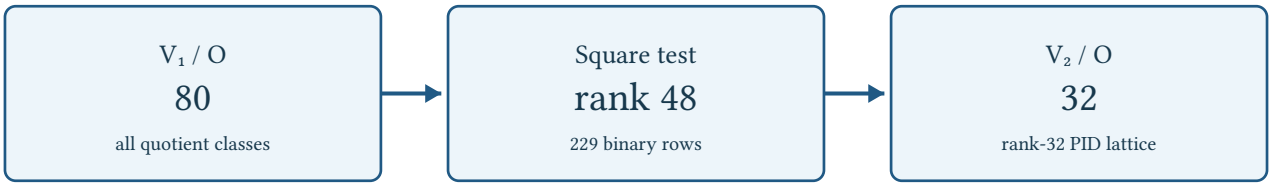
R66: second Frobenius lattice step

All 80 classes of $\frac{V_1}{O}$ are squared with the actual degree-32 source multiplication table. Complete denominator clearing gives 229 binary remainder rows of rank 48, hence 32 square survivors. Every remainder is reconstructed, every kernel product vanishes, and an altered kernel is rejected.

Second complete descending Frobenius step



Exact algebraic dimensions; run second-frobenius66-01. No timing comparison.

The full second PID lattice has rank 32. The quotient $\frac{V_2}{O}$ has binary dimension 32 and $\frac{V_1}{V_2}$ has dimension 48. Both module inclusions, both field-basis inverse products, old-order inclusion, and all 32 new basis squares back in V_1 pass. An altered inverse is rejected.

Execution and evidence

One native attempt finished with exit 0, the exact success marker, and no Singular error lines. Its UTC wall interval is 194 s, a receipt rather than a controlled performance comparison. Six historical source byte pins and six donor hashes are checked in results.json. The complete quotient, binary kernel, and polynomial/field lattice bundles are saved.

This is the second descending step. The next calculation tests all 32 classes of $\frac{V_2}{O}$ and continues until the lattice stabilizes. Actual proper branches and divisors follow that affine normalization.

The method follows Singh–Swanson, Theorem 1.1.